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Electric Line Deployed Shifting of Inflow Control Devices in a Herring Bone Completion and Ultra Sour, High Temperature Environment

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Abstract

As the depth and complexity of wells increase, the challenge to perform remedial interventions also increases. Extreme depth, tortuous trajectory wells with Herringbone completions in ultra-sour, high temperature environments are some of the most challenging wells in which to intervene. However, intervention is critical to ensure optimum production from each of the Herringbone laterals. This is achieved by mechanical manipulation to the desired state of the Inflow Control Devices (ICDs) in the mother-bore completion. Reaching the target depth and successfully performing such operations in these wells is extremely challenging as significant conveyance force is required in conjunction with high force, precise mechanical shifting of the ICDs. Traditional intervention operations with coiled tubing (CT) in such wells may not be possible due to lockup from frictional forces and the inability to apply the required shifting force even if the target can be reached.

In this paper we provide an overview of such challenges faced by an operator in the United Arab Emirates (UAE) and a wireline conveyed solution able to reach extreme depths to provide high force, precise, instrumented ICD manipulation. The approach required evaluating the history of mechanical interventions in the location to understand why previous attempts to shift ICDs failed. A lack of real-time downhole data, higher-than-expected shifting force, the presence of debris or corrosion, tortuous well trajectories and ultra sour environments were identified as contributing factors. A combination of technologies provided a comprehensive solution with the following methods:

- Production logging to obtain the inflow profile to identify the target ICDs to shift
- Conveyance and tension modelling of tortuous horizontal wells
- Using hydrogen-sulfide-resistant slim tractor and mechanical intervention tools with a high-strength electric-line cable
- Modelling and testing via surface integration tests (SITs) to shift the ICD and establish a baseline shifting signature

- Developing a slim-rotary mechanical-cleaning tool to remove debris from the ICDs
- Post-production logging to confirm successful shifting and the resulting production profile

The electric line-deployed mechanical cleaning and ICD-shifting tools enabled efficient closure of ten ICDs in an ultra-deep Herringbone completion well to allow targeted acid stimulation of the remaining open compartments and improve production. The lightweight deployment of this electric line solution significantly increased the efficiency of such an operation compared with traditional CT methods. The purely mechanical cleaning solution also allowed for localized cleaning of the target ICDs without requiring time consuming and expensive chemical treatments. The real-time instrumentation and downhole measurements integrated into the tools enabled precise manipulation of the ICDs during shifting with 0.1" accuracy of the displacement. The independent measurements of shifting force and displacement gave total confidence in successful shifting and helped to identify any sleeves which required more, or less, force to shift than expected. This successful deployment of electric line mechanical cleaning and ICD shifting tools opens opportunities for performing this type of intervention in even the most extreme wells.

Introduction

An operator in the UAE faced a long-term challenge to reliably manipulate ICDs in their extended reach drilling (ERD) wells. A significant number of wells across multiple fields are completed with ICDs in the lower completion. The upper completion in these wells typically comprises of slim 3-1/2" tubing with a lower completion of 4-1/2" pre-perforated liner. Swell packers set in the open hole section around the liner separate the well into compartments by isolating different production zones. The presence of ICDs within the compartments allows adjustment of the well inflow profile, by opening or closing ICDs as required, to achieve optimum production rates. During the lifetime of producing wells, as the zonal flow contributions and well performance changes, some zones will suffer from undesirable water breakthrough. In other cases, it may be required to isolate some of the zones to perform chemical stimulation on other zones with unsatisfactory production rates. To achieve the required objectives, in either case, requires a reliable solution to open or close the ICDs. Manipulation of ICDs in horizontal wells can be carried out using either coiled tubing or wireline. Historically, the success rate of shifting with conventional CT, where there is a lack of real-time feedback and the only data point would be surface tension, had been very low so the operator required an alternative method with a much higher likelihood of success.

In one of the highest producing fields for the operator, the wells are drilled from artificial islands with a very high density. To avoid collision between wells, and to maximize reservoir contact, the wells are designed with complex, tortuous trajectories. The resulting wells in this field are not only very deep, with extreme lateral lengths and tortuous trajectories, they also have a high temperature and significant presence of H₂S and CO₂. The combination of these factors results in an extremely challenging conveyance environment for any intervention. This is further compounded by the presence of internal diameter (ID) changes across the ICD assemblies, which means that any intervention tool and conveyance system must be able to navigate past these diameter changes deep into the well. Newer well designs aim to maximize production even further with the use of multi-lateral Herringbone completions. [Figure 1](#) shows the schematic of the lower completion of one such well. In this case the production from the Herringbone laterals is also controlled by the swell packer and ICD zonal isolation. The contribution from such laterals is a significant proportion of the total well production, so zonal isolation and control by opening or closing ICDs becomes even more critical. Such isolation may also allow for targeted acid stimulation of one or more laterals, by shutting off the ICDs of the other zones, to enhance well performance.

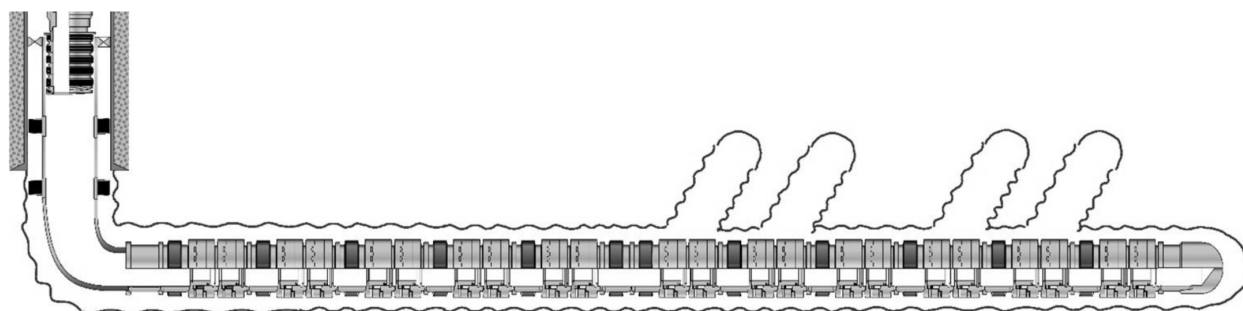


Figure 1—Herringbone Lower Completion with Zonal Flow Control Using ICDs and 4 Laterals

To determine the target ICDs for shifting, it is first necessary to evaluate the well performance in both shut-in and flowing conditions. The well performance must be assessed regularly to understand the inflow profile and to ensure that completion elements are performing as expected in this extreme, sour environment. This is critical to prevent any unexpected disruptions to well production in these key fields which may occur due to the risk of corrosion caused by the presence of high concentrations of H_2S and CO_2 . Wireline conveyance best practices for safe access to extreme wells in these fields to perform production logging have been well established on behalf of the operator with a proven track record of over 100 successful descents in this field (Varghese et al, 2023). These same practices are equally applicable to the deployment of wireline mechanical intervention tools as required for ICD shifting. The conveyance methodology is carried out in two stages, first the tools are lowered into the well on the wireline cable and descend by gravity in the vertical or low deviation section. Following this, the wireline tractor provides the necessary conveyance force to propel the intervention or logging tools to the target depth in the horizontal section, pulling the wireline cable behind. The wireline tractor is an electro-mechanical device that uses hydraulic pressure to open arms and provide sufficient traction force for the wheels to grip. The frictional drag of the cable in horizontal wells is significant and the tractor must generate sufficient pull force to overcome this and deliver the payload to the target depth. The wireline winch at surface is used to retrieve the tractor and downhole tools from the well by applying tension at surface when required. The personnel and equipment footprint of this wireline solution is relatively lightweight, while also removing risks such as helical lockup which may be observed with CT in such wells.

Historical challenges of ICD shifting for the operator resulted from the use of legacy CT or wireline systems with zero or minimal real-time feedback. As such it was not possible to be confident that ICDs had been shifted to the desired position and this left uncertainty following the intervention. In some cases, the presence of debris within the ICD assembly made it difficult for the shifting tool to latch onto the profile of the sliding sleeve or resulted in very high force to perform the manipulation. Therefore, a three-pronged solution was required; firstly, removing the uncertainty of the shifting operation success, secondly, removing any potential debris, and thirdly, the ability to reliably apply very high shifting forces. A wireline mechanical intervention platform with integrated instrumentation, providing real-time feedback and precise control of downhole operations, was an optimum solution to overcome such challenges. The presence of multiple, independent sensors within the downhole tools could give a precise indication of the force required to move the sleeve as well as the total displacement which could be matched accurately to the ICD technical schematics. A custom, slim outer diameter (OD), expandable, high-speed rotary mechanical cleaning tool could perform localized cleaning to remove any debris from the ICD assemblies. Finally, a high-force downhole shifting tool would allow for ICD manipulation at the required forces. When shifting on CT the push or pull force is applied from surface, and therefore as the well depth or complexity increases, the available manipulation force decreases as the frictional force acting on the coil is greater. The wireline shifting tool utilizes a downhole anchor and high-force linear actuator, or stroker, to apply thousands of

pounds of shifting force consistently, regardless of the depth. This paper discusses the conveyance and intervention challenges in detail, with a tailored combination of solutions that were used to overcome them.

Intervention Challenges in Sour, Tortuous, ERD Wells

Extreme Trajectories and Accessibility

The traditional method of shifting ICDs was using either slickline or conventional coiled tubing. Slickline operations are suitable for lower deviation wells and provide an efficient, low-cost solution. Once in position, a jarring action from surface will provide the required force to shift the ICD. As the well deviation increases, the ability to provide sufficient jarring force downhole diminishes and eventually with very high deviation it will not be possible to even reach the target. Deeper, deviated wells will also typically have multiple ICDs with the same shifting profile, and this can present an additional challenge. As the mechanical shifting tools are not selective, they will engage with every ICD that is passed or may only be able to engage with the topmost ICD where it is unable to pass any deeper due to the no-go profile of the shifting tool. When an operator wishes to selectively close different zones in a well this purely mechanical approach becomes incapable of achieving the objective of selectively shutting off specific targets. To overcome the challenge of deviated wells, coiled tubing was an alternative conveyance solution to perform shifting operations. Coiled tubing is well-suited to deviated or horizontal well applications, with push force being applied from surface to feed the coil into the well. This conveyance method has limitations as the well gets deeper and helical lock-up may be observed, where the frictional force encountered by the string running on the wellbore tubular reaches a critical point. Although more tubing may be injected into the wellbore, the end of the tool string cannot be moved farther into the wellbore. This limitation can be overcome with the use of downhole agitators in the bottomhole assembly (BHA) which provide axial oscillations along the length of the string to reduce downhole friction and extend the point of lockup. Despite the use of such techniques, in very deep, tortuous wells with extreme friction it may still be impossible to reach the target depth with coiled tubing.

There are multiple factors to overcome for successful conveyance in these extreme wells. The extended laterals, tortuous trajectories, high temperatures and presence of H_2S and CO_2 combine to create very challenging conditions for conventional conveyance solutions. The presence of sharp turns in the well-trajectory creates capstan points where friction is much higher. [Figure 2](#) shows the 3D model of the well profile from the case study covered subsequently in this paper. The trajectory is observed to change simultaneously in all three axes, and it is this tortuous trajectory that results in a rapid increase in friction deeper in the well. The frictional drag on a coiled tubing or wireline cable compared to a more typical trajectory adds considerably to the tension during running in hole (RIH). In the latter case, a wireline deployed tractor provides the conveyance force necessary to reach the target depth by pulling the cable with sufficient force to overcome the observed friction while also pushing the payload of logging or intervention tools below itself.

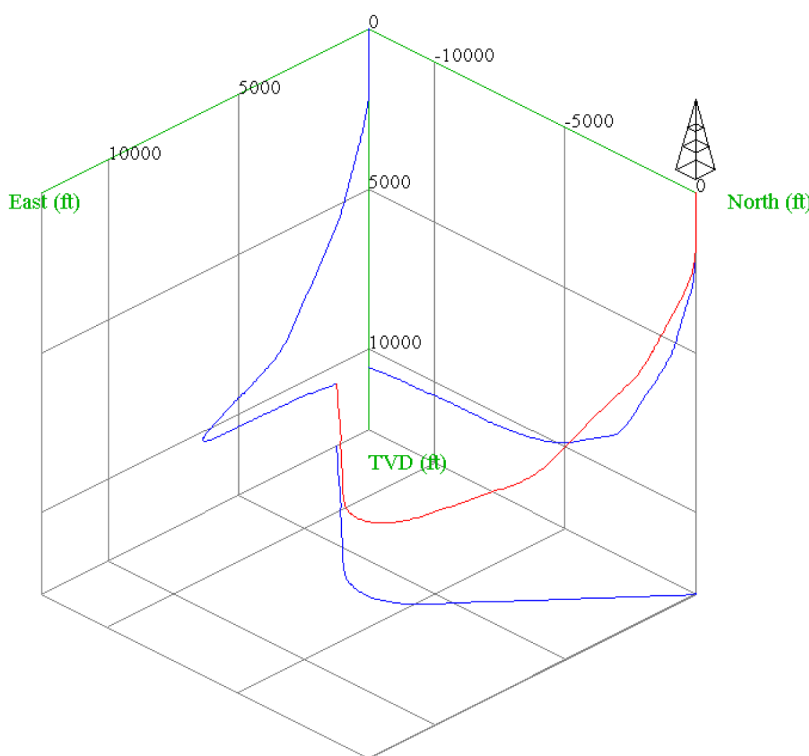


Figure 2—Trajectory and profile of ERD tortuous well

Due to the complex well profile and trajectory, the difficulty to install the completion also increases. During installation, the tubing may buckle, and it is not always possible to precisely land the end of the tubing at the exact target depth. This means that in some instances there can be a gap between the upper and lower completions, which adds to the conveyance challenge. While simultaneously navigating such gaps, it will usually be the case that there are multiple ID changes to pass in the completion. The operation of a wireline tractor requires the application of hydraulic force to open the arms to provide traction force to the wheels. When these tractor arms are passing through any gaps or ID changes in the completion it may cause the wheels to hang up or stall. As such, it will be required to close some tractor sections while they pass through, relying on the conveyance force provided by the remaining active sections. In a traditional wireline tractor design 50%, or one out of two deployed tractor sets, must be completely closed during navigation so there is a significant reduction in conveyance force. This can result in there being insufficient conveyance force available to continue the descent. As the depth of the well increases, the frictional cable drag increases and so this reduction in conveyance force becomes more significant. To continue to RIH, overcoming this cable friction, will require the tractor arm opening force to increase to provide additional traction force on the wheels. As the opening force of the arms increases the system becomes more rigid, meaning that the tractor is less likely to pass any ID changes without stalling. The ability to precisely control the tractor speed and arm force for individual drive sections becomes invaluable, therefore, to navigate the ID changes in ERD wells. Figure 3 shows a wheel-based wireline tractor with two drive sections, offset by 90 degrees, visible.



Figure 3—Intelligent, Instrumented, Sour Well Wireline Tractor with Individual Arm Navigation Capability

Resources Footprint and Efficiency

The wellsite location for these intervention operations is typically on an artificial island, constructed offshore to provide a facility from which to drill complex, ERD wells. Such locations are constrained in terms of available accommodation space for personnel and the area around the wellhead to set up the required equipment. Performing interventions on coiled tubing requires a large team of personnel to operate, due to the complex nature of the surface equipment, which may not always be possible to accommodate depending on other ongoing activities at the wellsite. The significantly larger equipment footprint may also pose additional operational challenges. Accessibility to some of the wellheads is limited and larger equipment may be difficult to move into the required position for the intervention. Large volumes of fluids and pumping capabilities are necessary for many of the coiled tubing functions, and this contributes greatly to the equipment footprint. The setup time on the well for this large, complex equipment package will take multiple days and be heavily weather-dependent due to the significant weight of the individual pieces of equipment. The time taken to rig-down the equipment following completion of the operation is equally long. Overall therefore, the large and complex nature of the coiled tubing package makes it an inefficient option compared with lighter weight alternatives such as wireline or slickline. Efficiency is a critical factor for the operator's decision-making process for interventions as a faster operation means that a well only needs to be offline for a shorter duration before production can resume, and such interventions can more easily be fit into a tight schedule of required works.

Lack of Real-time Measurements

Performing interventions with legacy methods of slickline or conventional coiled tubing proved challenging as there was zero real-time feedback from downhole. With the application of surface force, tensile or compressive, for coiled tubing or jarring action for slickline it is attempted to shift the ICD into the desired position. An indication at surface can only imply that force has likely been applied to the ICD sleeve but there is no way to know if it has been successfully shifted. Specifically focusing on interventions in horizontal wells, therefore discounting slickline due to being unable to reach the target depth, the coiled tubing injector head applies the push or pull force on the coil. This force is transmitted through the entire string down to the shifting tool and as such a significant portion of the surface-applied force must overcome the frictional drag of the coil to then act on the ICD sleeve. Distinguishing between the force required to overcome this drag on the coil and the actual force being applied to shift the valve is very difficult. This may be further compounded depending on the type of sleeve to be shifted. Some sleeves require only a few hundred pounds of force to move a short distance between open and closed positions; this will be such a small proportion of the total force applied at surface that it will be completely masked. In other cases, sleeves may require significantly higher forces than expected to shift. This could be due to corrosion or the presence of debris across the ICD assembly mechanically locking the sleeve. The shifting tool may apply many thousands of pounds of force on the sleeve, but due to the wellbore conditions it does not move as the corrosion or debris has prevented it from doing so. With only surface tension feedback, it may be understood that the required force had been applied to the ICD and then assumed it has moved to the desired position. At these more extreme shifting forces, there is also the risk of accidental disengagement of the shifting tool. Therefore, it could arise that the required force was applied, however the sleeve didn't move, and then the shifting tool accidentally disengaged.

These scenarios have been observed in many instances, resulting in an expectation that a shifting operation has been completed successfully only to later discover from subsequent production logging that the sleeves are still in the incorrect position. Advances in coiled tubing and wireline intervention tools marginally improved this situation. The addition of a downhole tension sensor increased confidence that the force was being transmitted and applied to the ICD. However, without also being able to observe the linear displacement of the sleeve during the shifting operation the uncertainty remains as to the final position. Considering the time taken not only to perform the intervention, but also the subsequent production logging and surface well tests for validation of the well performance, this uncertainty has a massive impact on the confidence of the operator to perform such operations.

High Shifting Force

Typically, the force required to manipulate ICDs is in the range from 1,000 to 3,000lbf per manufacturer specifications. There are some exceptions, but this represents the most common shifting force range of sliding sleeves. If shifting an ICD in a deviated well with coiled tubing, the force applied from surface at the injector head must be transmitted downhole through the entire string to act on the shifting tool and shift the sleeve. The tortuous trajectories and extended reach laterals of these wells create capstan points that significantly increase the frictional drag acting on the coiled tubing string, the same as observed for conveyance. Once deep into the horizontal section of the well, typically after a sharp turn to the side, this friction force increases at a non-linear rate. In such scenarios, the frictional force acting on the coiled tubing string may be so great that there is little to no force left to push or pull with the shifting tool once the target is reached. In typical wells of this field, the available shifting force can be reduced to as little as 1,000 to 1,500lbf which limits the ability of coiled tubing to shift a wide variety of ICD types.

The risk of corrosion, debris or scale deposition in such wells is relatively high. The presence of such corrosion at the interface between the sleeve and ICD assembly, or if any debris build-up is present as a physical barrier to the sleeves' movement, can result in the shifting force increasing to over 10,000lbf. In such cases the limited available coiled tubing shifting force downhole will be several times less than the required shifting force of the ICD and it will not be possible to perform the intervention. The wireline deployed intervention systems de-couple the cable from the shifting operation and therefore negate the significant frictional force effect in these wells. This is done via the use of a downhole anchoring system to lock the BHA in place once the ICD is engaged. Once locked in position, a hydraulic piston will provide the extension or retraction force to manipulate the sleeve. The typical wireline intervention systems available to perform such operations may provide the required shifting force of over 10,000lbf but have a limited expansion ratio. Therefore, a slim diameter shifting tool which may pass the minimum well restriction in the 3-1/2" upper completion, may not also be able to operate in the larger lower completion to shift the ICDs.

Presence of Debris or Corrosion

In addition to the potential of increasing the force required to shift the ICD sleeves, the presence of any debris or corrosion can make engaging with the shifting profile of the sliding sleeve more difficult. The cross-sectional area of engagement between the shifting tool 'key' and the sleeve shifting profile is small, as seen below in Figure 4, and typically relies on a hard 'shoulder' with a 90-degree angle to allow a firm interface to apply the required high shifting forces over this small area. The example in this figure is for a wireline shifting tool, but the situation is the same for a flow-activated shifting tool used with coiled tubing. There is no way to have a large cross-sectional area of engagement as this would require the ICD sleeve to protrude into the wellbore further, increasing the risk of accidental latching during any downhole operation, and potentially interfering with well production. As this protrusion is small, any debris can prevent clear access to the shoulder of the ICD sleeve shifting profile and make it impossible to latch at all. Even if it is possible to get an initial latch at low forces when locating the shifting profile, once the high tensile or compressive forces are applied to perform the shifting operation the tool may accidentally disengage. Without any downhole

feedback, such accidental disengagements may appear identical to an expected automatic disengagement which occurs upon completion of the shift. Additionally, if such accidental disengagements happen under high forces, this can cause damage to the shifting key and round off the hard shoulder which will make it even more difficult to subsequently latch or shift a sliding sleeve.

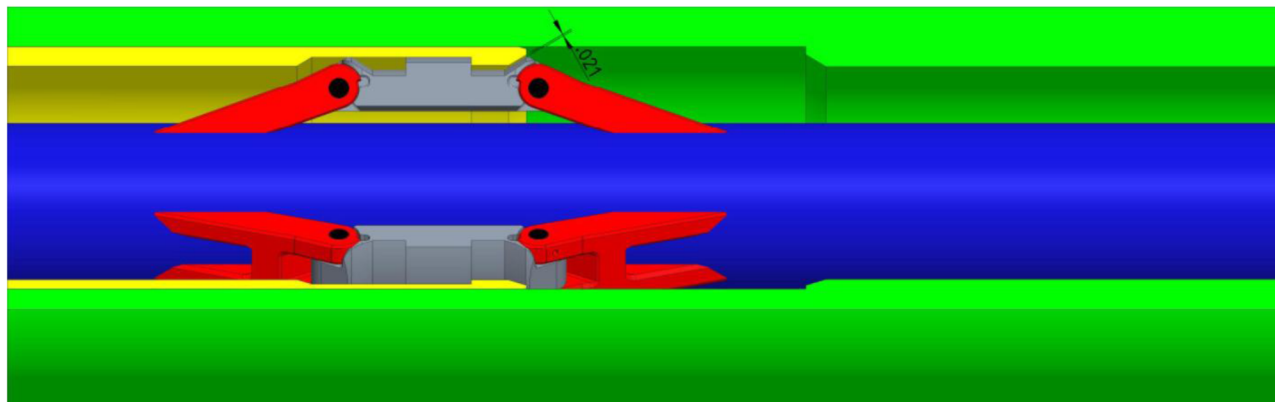


Figure 4—Shifting Key (grey) Engagement with Shifting Profile of the ICD Sleeve (yellow)

Prior to any ICD shifting operation, if the well has a history of debris build-up or there is any expectation of the presence of scale or corrosion, it is first required to perform a cleaning operation to ensure it will be possible to manipulate the sleeves as required. Historically, the only solution for this would be to deploy coiled tubing and use a chemical-based cleaning tool. This utilizes a jetting BHA that directs the fluid, pumped down through the coiled tubing assembly, at a high speed towards the internal face of the completion. This jetting can either be from a stationary or rotating orifice to clean effectively. Such solutions are highly effective at performing a full wellbore clean-out, breaking down any viscous fluid, and circulating debris back to surface. However, they require a significant amount of time to perform as well as a large volume of chemicals to be pumped into the well and then displaced back to surface. If the primary operational objective is the successful shifting of the ICDs, then such a complete and heavy-duty wellbore clean-out may be excessive and a lighter-weight, more efficient approach which just targets the zones across the ICD assemblies would greatly increase overall efficiency for the operator.

Extreme Sour Environment

The ultra-sour environment in this field, with extreme concentrations of H_2S and CO_2 , poses a significant risk to coiled tubing, wireline cables, wireline tractors and any intervention tools in the form of hydrogen embrittlement, sulfide stress or corrosion. Non-sour coiled tubing equipment would not be considered for such operations due to the high risk of a well-control incident and only specifically engineered sour service equipment would be viable. Standard wireline equipment, designed for non-sour environments, may survive for a limited number of operations without immediate failure but it would rapidly become prohibitive to operate in this way as it would require frequent replacement of many components. Wireline tractor and intervention tools have many moving parts and dynamic seals which can allow the ingress of a small volume of gas inside the hydraulic system, and this makes them especially vulnerable to the sour environment. The accumulation of this gas inside the tools with a long exposure period will potentially cause catastrophic equipment failure. It is imperative to select a cable, tractor and intervention tools that are specifically designed to operate in such an environment and to follow a meticulous maintenance regime to ensure long-term reliability of the intervention service.

Intervention Solutions for Sour, Tortuous, ERD Wells

High Power Tractor and Torque Balanced Cable

The slim diameter, intelligent, sour-service tractor utilized for this operation has a modular design which can be run in different configurations, varying the number and spacing of the tractor drives, depending on the conveyance and navigation requirements of the well. Increasing the number of tractor drives will provide additional conveyance force to reach deeper targets, but it will also result in greater electrical power consumption. Therefore, a balance between the required conveyance force and the electrical power constraints of the wireline system must be achieved. The optimum configuration for the slim tractor in this operation was a '3 + 3 tandem' configuration, comprising of two sets of tractors with three drives each. The unique ability of this tractor to individually open or close single drives, rather than entire drive sets as in other systems, while navigating restrictions meant that there was a significant reduction in the loss of conveyance force. Rather than losing 50% of the available force from closing an entire set of drives, closing just one out of 6 drives results in around 17% force reduction. The operators' wells have ICDs deep in the lower completion, where the required conveyance force is high, and so this ability to close just a single drive while navigating is not just useful but becomes necessary. Figure 5 compares the two scenarios of navigating past an ICD, one closing an entire tractor set and one closing just a single drive at a time. In cases where the distance between the tractor drives matches multiple ID changes in a well, which would make navigation more difficult, the modular nature of this tractor allows the drive spacing to be adjusted. Altering the spacing ensures that the tractor drives will be positioned such that only one of them passes a restriction at a time. Accurate modelling of well restriction navigation to avoid this issue is performed for each well during the job design stage in addition to the standard tension simulations.

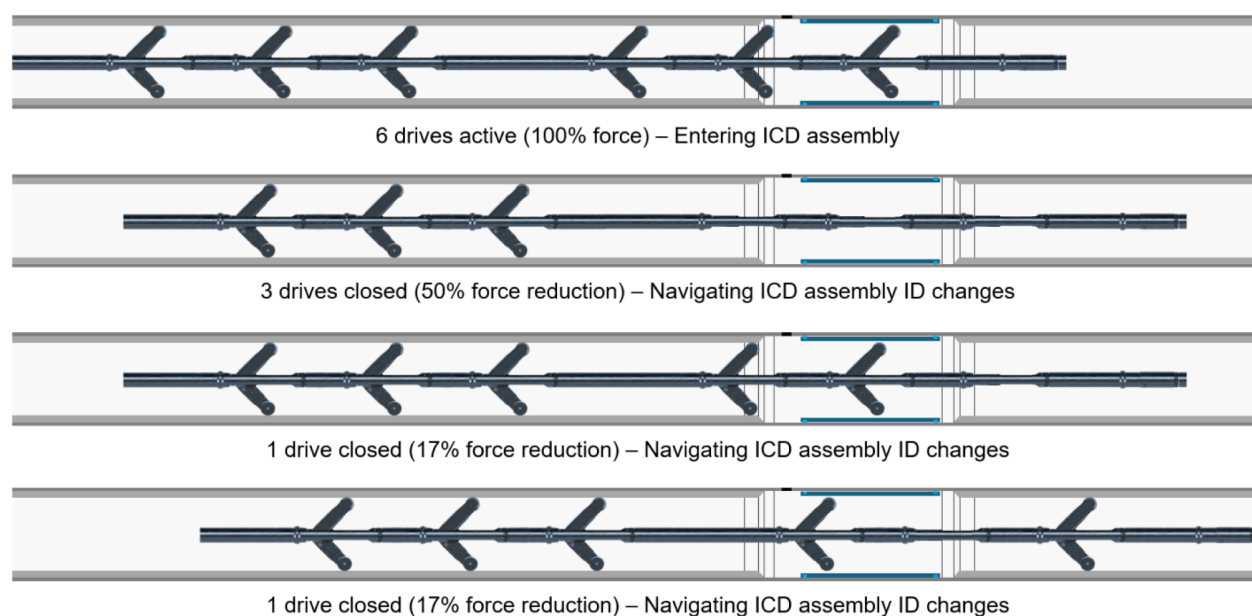


Figure 5—Available tractor force during RIH and restriction navigation with either three or one drive closed

The intelligent, instrumented design of the tractor allows for a precise understanding of what is occurring downhole during conveyance, which is especially critical in ERD wells or when navigating restrictions. The ability to monitor and react correctly, in real-time, ensures that appropriate decisions are taken to optimize the tractor's performance throughout any operational challenges. The tractor responds to commands sent from surface to initiate the operational steps, but then the system is self-regulated downhole. The tractor firmware logic responds to the measurements taken from the sensors to react and send signals to the

electronic systems as required. This automation allows for a much faster response to the downhole operating conditions compared with a surface-controlled system of a non-instrumented, non-intelligent tractor. User intervention is possible at any time for the engineer to adjust parameters or send new commands to the tractor when required. Measurements such as casing collar locator (CCL), downhole tension and compression give a clear indication of smooth tractor progression into the well. Integrated pressure sensors inside the tractor hydraulic system and three-dimensional accelerometers can clearly identify when a drive section is passing a restriction. This comprehensive suite of downhole data is transmitted to the surface real-time via a high bandwidth telemetry system, viewed and controlled in a graphical interface which allows for a clear understanding of the status of the operation and any required restriction navigation with the tractor.

Reaching the target depth is just the first part of the conveyance challenge, it is also necessary to safely retrieve the downhole tools to surface with the wireline cable once the operation is complete. Therefore, wireline cable design and construction play a significant role in overcoming both aspects of the conveyance challenge. While running in hole to the target depth, the cable should provide the necessary high electrical power for tractor operation, have low weight and low friction to enable ERD well access, be able to survive in the harsh sour environment, and withstand any effect of accumulated torque during the operation. When retrieving the tools to surface, the cable should also be sufficiently strong, with a high enough safe working load (SWL), to apply enough downhole tension at the tools to pull them out of the well. The demanding cable design and technical criteria for such operations mean that there are very few options that exist to perform these interventions successfully. The state-of-the-art cable technology and conveyance techniques are now well established for such wells for the operator to perform wireline interventions (Varghese et al, 2023). The optimum cable for these operations with a flawless track record has a high electrical power rating, is H₂S and CO₂ resistant, has a high strength-to-weight ratio with sufficient SWL capacity, and a unique torque balanced design. This design has a polymer filling between the armors, as well as larger diameter inner armors, which prevents torque accumulation, therefore operations become both safer and more efficient. The cross section of this torque-balanced wireline cable can be seen in Figure 6.

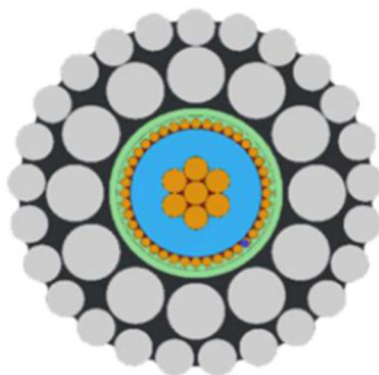


Figure 6—Cross-section of torque-balanced wireline cable with polymer fill between armors

Lightweight, Efficient Deployment

To support the operator in managing simultaneous operations on their artificial islands with constrained accommodation space and limited area to set up equipment around the wellhead, a lightweight solution was required. The deployment of intervention solutions on wireline, combined with a wireline tractor, allows for a minimized footprint both in terms of equipment and personnel. The wireline unit utilized for interventions is identical to a standard unit used for conventional wireline operations such as logging. As such, complete deployment requires just a single wireline unit or truck and a control panel to operate the pressure control equipment. Compared with coiled tubing interventions which would require multiple large units set up on surface, and a much larger installation of equipment on the wellhead, this is a significant reduction in

equipment footprint. The reduced number of pieces of equipment, and their much lighter weight, reduces setup time and risk of positioning equipment during job preparation and post-job. The crew size to support a wireline operation is typically less than half the size of a coiled tubing crew as there is less surface equipment to manage. As such, the overall operation time of wireline operations is significantly reduced due to both the lightweight surface setup and reduced complexity of pressure testing, combined with faster deployment and retrieval of the downhole tools. In addition to the operational efficiency, wireline operations typically also result in around 60% reduction in CO₂ emissions compared with coiled tubing. With the ability to perform a wide range of interventions on wireline, including mechanical cleaning and ICD shifting, the operator can more easily schedule such work in a proactive and timely manner for the most effective results.

Instrumented Intervention Platform

Prior to the mechanical intervention operation, typically it must first be ascertained which zones, and therefore which corresponding ICDs, need to be manipulated. To determine this, production logging data must be acquired in these horizontal wells with three-phase flow. Acquisition of such data requires logging tools specifically designed for multi-phase flow which may separate into individual phases across the wellbore. The tool selected for this data collection was an advanced, multi-array production tool which is qualified for this harsh environment with a significant concentration of H₂S and CO₂. This tool comprises of an array of five high-sensitivity flowmeters, which determine the fluid velocity distribution across the cross-section of the wellbore. In addition to the flowmeters, the tool houses an array of 6 water-holdup sensors, with resistive-based measurements, and 6 gas-holdup sensors, with optical-based measurements. These sensors accurately measure the three-phase hold-ups of gas, oil and water across the wellbore cross-section. The optical sensors may also identify the initial entry of gas bubbles inside the wellbore during flowing conditions to confirm the bubble point pressure. The arm which comprises this sensor and flowmeter array has an integrated single-axis caliper and orientation sensor to determine the wellbore ID. A schematic of this compact, integrated, advanced production logging tool may be seen in [Figure 7](#). Additional modules as part of the logging BHA provide other standard measurements such as high-resolution pressure, temperature and deviation to supplement the production logging data. Performing such operations via wireline significantly reduces the choking effect which may occur when coiled tubing is deployed in a slim completion and allows the operator to acquire more representative production logging data at higher flow rates.



Figure 7—Advanced multi-array production logging tool to determine three-phase fluid velocity and hold-up

Following acquisition and interpretation of the production logging data, the target ICDs for manipulation are confirmed. The historical challenges of performing these ICD manipulations are well documented. Without real-time feedback and downhole data, the operator has been unable to consistently and confidently shift ICDs to the desired position. The only way to overcome this, and to be able to efficiently perform intervention operations with a high degree of confidence, backed by downhole measurements, is to utilize intervention tools that are fully instrumented.

The intervention tools selected for this operation are part of a suite of fully instrumented tools, equipped with an array of downhole sensors. This replaces the legacy approach of attempting to provide interpretation and confirm successful intervention based on a very limited set of measurements, potentially surface tension only. The downhole sensors acquire the data, and the tool electronics transmit this in real-time to surface via the wireline cable to understand the status of the operation. Each action or movement of the tool has a corresponding sensor monitoring it, and so any desired intervention can be verified and adjusted by the engineer with a graphical interface. This allows the engineer to take the correct actions throughout the

process and to react to any unexpected outcome. This could include for example, higher or lower than expected shifting force or accidental disengagement from the sleeve. In the case of shifting ICDs, applying the expected force may not actually result in any movement of the sleeve depending on the sleeve and downhole conditions. The integration of a millimeter precision linear displacement sensor in the shifting tool provides an additional reference point for the manipulation. The distance of sleeve travel measured by the sensor can be verified versus the technical schematic of the ICD assembly and compared with the observations made during the pre-job SIT. The value of these integrated sensors and the ability to successfully shift, and verify with direct measurements, a wide range of ICDs in ERD interventions has been well demonstrated by this platform (Mauchien et al, 2021). The shifting tool utilized in this operation, as seen in Figure 8, comprised of three main modules, a downhole anchor which locks the tool in position to perform the intervention, a linear actuator which provides the push or pull force to manipulate the sleeve, and the shifting tool which interfaces with the sleeve profile during the shifting operation.



Figure 8—Wireline Conveyed Instrumented, Precision, Universal Shifting Tool

Each module can be individually or simultaneously controlled as required, and contains independent sensors which contribute valuable data to the successful shifting operation; the anchor force and internal diameter readings ensure that the tool is located where expected and is not slipping, the linear actuator force and displacement sensors confirm the successful shift, and the hydraulic opening pressure of the shifting tool provides a clear picture of the shifting key engagement and disengagement with the sleeve. The combination of all these sensors provides a comprehensive data set for real-time analysis of every step of the operation and completely removes any doubt of the final position of the sleeve. With such certainty, the operator can confidently perform the required ICD manipulations and efficiently move to the next operation. A comparison of the real-time measurements available from the different shifting options is seen below in Table 1.

Table 1—Real-time measurement comparison of various shifting methods

Shifting Option	Feedback and Measurements					
	Surface Tension	Head Tension	Linear Position	Shifting Pressure Signature	Downhole Pressure	Sleeve Diagnosis
Slickline	Yes	No	No	No	No	No
Coiled Tubing	Yes	No	No	No	No	No
eCoil	Yes	Yes	No	No	Yes	No
Instrumented Wireline	Yes	Yes	Yes	Yes	Yes	Yes

Slim Diameter High Force Shifting Tool

All intervention tools deployed in these wells must be of a slim diameter to pass through the narrow restrictions of the upper completion. In general, this necessitates tool diameters to be less than 2.5" to have sufficient clearance to safely pass these restrictions. The interventions typically take place in the lower completion where there is a 4-1/2" liner with an internal diameter close to 4". To manipulate the ICDs in the lower completion the shifting tool must have a sufficient expansion ratio to pass through the restrictions then open wide enough to anchor in the tubing and engage with the shifting profile of the sleeve. The shifting tool selected for this operation, as seen in [Figure 8](#), has a slim OD of 2-1/8" but an extreme expansion ratio which allows it to operate up to 5" internal diameter completions ([Mauchien et al, 2019](#)). This can be further augmented for larger completions with expansion kits if required.

The typical force required to shift ICDs during the interventions is expected to be less than 3,000lbf for the majority of ICDs, as per the manufacturer specification. However, historical experience both for the operator and globally has shown that in some cases the required shifting force can exceed 10,000lbf. As such, not only must the shifting tool be able to provide the required expansion ratio for the operation, but it must also be able to do so while simultaneously providing very high shifting forces at extreme depths. The use of a downhole hydraulically controlled anchor and linear actuator completely de-couples the application of force from surface and as such the available force to shift the ICD is independent of depth. The 2-1/8" OD shifting tool deployed for this operation had the capability to apply up to 16,000lbf of force to manipulate the ICD sleeves which provides a significant excess force contingency to overcome a partially stuck sleeve. The higher-than-expected shifting force may be due to the presence of debris or scale in the ICD assembly. In some cases, a direct application of high force is still insufficient to cause the sleeve to move completely. When this is diagnosed, via the downhole sensors of the instrumented shifting tool, the approach may instead be to cycle the sleeve up and down by very small movements, confirmed by the linear displacement measurements, to break this debris and allow a complete shift. The high expansion ratio combined with high displacement force of the instrumented shifting tool enables the operator to successfully shift ICDs even with very high force requirements at extreme depths in the well.

Mechanical Cleaning Tool

The potential presence of any debris or corrosion developing around the ICD sleeve may result in a significant increase in the force required to shift or increase the likelihood of accidental disengagement with the shifting profile. Simply attempting to apply high force to shift the sleeves in such cases often proves unsuccessful. For situations where there is a track record or expectation of debris presence in the well, the operator required an option to remove it prior to the shifting operation. Historically this may be done with the use of a coiled tubing deployment that utilizes chemicals jetted at high velocities through the wellbore to provide a complete clean-out. This requires the setting up of a large package of equipment and dealing with the chemicals once returned to the surface. This approach is relatively time consuming which decreases the overall operational efficiency significantly. The fundamental requirement in this operation was to perform a targeted cleaning of only the ICDs prior to shifting in an efficient manner and without any chemicals. Wireline intervention capabilities have broadened to offer milling and debris collection for wellbore cleanout. A standard milling option to remove debris would be limited in its cleaning effectiveness as the mill bit must pass through the smaller ID of the upper completion and then it may not expand all the way to the internal face of the larger ID lower completion to remove debris completely. Furthermore, even if a mill bit had an expansion capability to reach the required diameter, the variable ID and delicate nature of the ICD assemblies would potentially be damaged with this approach regardless.

Building on the foundation of the instrumented wireline milling tool, a high-speed expandable cleaning brush was developed for the operator as a chemical-free cleaning solution. In this application, the rotational speed was increased to several hundred revolutions per minute (RPM) though the use of an alternative gearing configuration. In place of the fixed diameter milling bit, an expandable brush with a linkage

mechanism was developed with a high expansion ratio as seen in Figure 9 below. With a running OD of 2.2" the brush could effectively clean up to a 5.5" completion ID. The flexibility of the brush bristles compared with a milling bit also removed the risk of any damage to delicate completion components such as ICD assemblies. The inbuilt instrumentation of the milling platform at its core allowed for real time monitoring of the rotational speed and torque which allowed an understanding of effective cleaning of the ICDs. The tool had the ability to automatically self-regulate within set parameters of rotational speed and effective weight-on-bit (WOB), provided from the tractor push force, input into the software. This automation, originally developed for milling applications, allows the system to precisely control and adjust the tractor push force to maintain a higher rate of cleaning for a more efficient operation with less stalls and a lower risk of becoming stuck. (Wiesenborn et al, 2019)

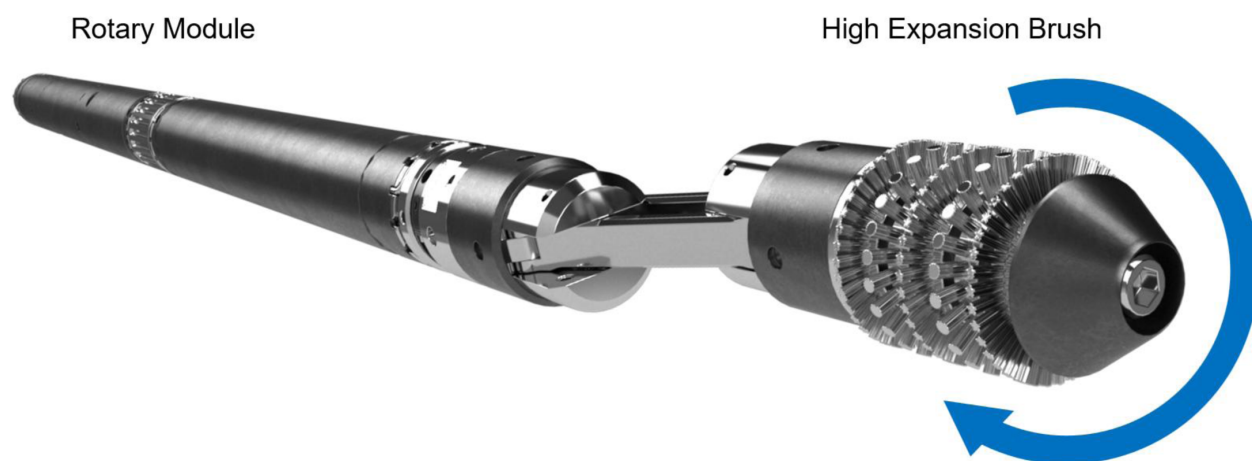


Figure 9—Wireline Conveyed Rotary Mechanical Cleaning Tool

Sour Service Equipment

The harsh well environment required a comprehensive set of equipment which must all endure such conditions, combined with a strict maintenance regime to ensure uninterrupted service delivery for the operator. All downhole tool components and the wireline cable were carefully selected to meet these stringent requirements. The tractor and intervention tool housings are made of Inconel, Hastelloy, Titanium and other NACE compliant materials. Similarly, the torque-balanced, high-strength cable selected for this operation was also fabricated from H_2S and CO_2 resistant alloy to be able to operate in this environment. The operator typically required multiple runs or multiple operations to be completed back-to-back and therefore the selected equipment must be able to meet this requirement. Due to the nature of wireline tractors and mechanical intervention tools, the presence of dynamic seals means that small amounts of H_2S and CO_2 will enter their hydraulic systems during operations. To flush out this gas to prevent internal corrosion and to provide the most efficient maintenance possible to sustain continuous operations, a specially designed mobile maintenance laboratory with a highly specialized technician is always present. This allows for a rapid turnaround of tools for uninterrupted, multi-well operations without demobilizing equipment.

The challenges associated with accessing these tortuous, ERD wells in a sour environment and performing mechanical interventions with wireline are summarized below in Table 2. Only with the unique combination of sour service intelligent, instrumented, modular tractor and intervention tools combined with a high strength, high power, torque balanced cable is it possible to overcome these challenges and reliably perform the required interventions for the operator.

Table 2—Summary of conveyance challenges in tortuous, ERD, sour wells

Intervention Challenge	Solution
Risk of corrosion due to presence of H ₂ S and CO ₂	Sour service equipment and strict maintenance regime
High conveyance force to reach target depth	High-force, intelligent wireline tractor
Complex navigation through ICDs throughout the well	Modular tractor with variable drive quantity and spacing
High cable tension due to friction in long lateral	High strength-to-weight ratio cable with >10,000lbf SWL
Significant risk of cable armor damage due to torque	Torque balanced cable combined with swivel
Limited power delivery of sour-service wireline cable	Cable with high power delivery capability >4A
ERD wells and very long duration operations	Light weight wireline surface equipment deployment
Limited accommodation space and tight schedule	Minimal wireline crew size and efficient operation
Lack of real-time measurements and uncertainty	Fully instrumented intervention platform
High shifting force to manipulate ICDs	Slim OD, high expansion ratio, high force shifting tool
Potential presence of debris or corrosion	Expandable, high speed, rotary mechanical cleaning tool

Operational Planning

Meticulous planning of the operation was performed collectively by the operator and service company. A wide range of contingency scenarios were considered as part of risk assessment and mitigation planning. As the well to be intervened was extremely deep and tortuous, with high concentrations of H₂S and CO₂, it was imperative to design the operation to be as efficient as possible to minimize the downhole exposure of the tools. For the acquisition of production logging data, the tools would be conveyed under flowing conditions to allow for stabilization of flow conditions in advance to yield higher quality data in a timely manner. Using this approach, both flowing and shut-in data can be acquired with a single descent into the well. For the intervention operation, mechanical clean-out of the target ICDs would be done in parallel with running in hole so that the cleaning could all be completed on the way down with additional descents. Finally, for the ICD shifting the optimum operational plan was to reach the deepest target ICD to be shifted and then consecutively shift the subsequent target ICDs while pulling out of hole.

Contingency Planning

A summary of the various contingencies planned and implemented for the operation are as follows:

1. **Stuck Tools** – Once initial attempts to free the tool were taken, the next steps would include measures such as flowing the well or pumping friction reducer to try to change well conditions prior to pulling at the maximum safe limit if the cable
2. **Release Device** – If no measures were successful in releasing the tools, then an emergency release device within the BHA could be used to disconnect and safely retrieve the cable to surface. The release device selected for this operation was electrically operated and could be activated through a real-time command from surface or, as a contingency, through a battery powered downhole timer. The timer can be set and reset from surface during the operation, but in the case of loss of cable integrity and communication with the tools, then the timer would automatically activate the release device once the pre-set timer elapsed.
3. **Stuck Cable** – If it was not possible to free the stuck tools and the release device failed to disengage, a pump-down cable cutter could be deployed from surface to reach the logging head and sever the cable which could then be retrieved to surface.
4. **Fishing** – Contingencies of mechanical fishing tools compatible with coiled tubing, wireline intervention and slickline BHAs were tested to ensure consistent engagement with the fishing profile of the emergency release device.

Simulations and Equipment Testing

As part of the operational planning, detailed simulations and equipment testing were conducted. The shifting profile of each type of ICD sleeve can be different, along with various unique ID changes and shoulders encountered when passing through the assembly. To ensure that the shifting key would engage with a sufficient cross-sectional area of the profile, and to check for any potential unintended hang-ups on any other profiles, the entire latching and shifting process is simulated in 3D as seen in Figure 10. This simulation can determine with a high degree of accuracy the likelihood of success of the entire process of latching the profile, shifting the sleeve to the desired position and the disengagement behavior of the shifting tool upon completion of the shift to remove any doubt.

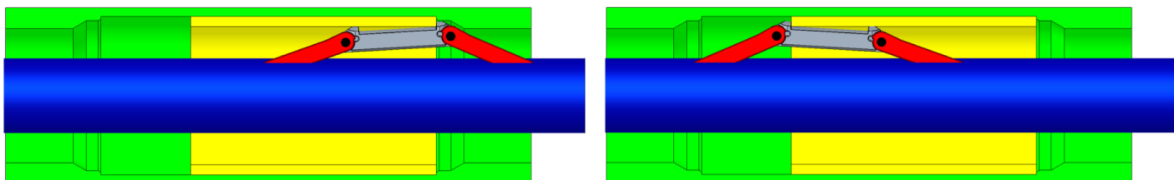


Figure 10—3D Simulation of Shifting Tool Engagement with ICD Sleeve

Following comprehensive 3D modelling and simulation, a custom shifting key was manufactured for the operation. Building on the expected behavior from this simulation, a full equipment package to replicate downhole conditions was set up in the workshop for an SIT. Multiple ICD assemblies were tested one-by-one, connected to tubing joints within which the wireline tools were installed. The full wireline BHA with tractor and intervention tools were connected to allow simulation of the downhole operation as closely as possible. The SIT steps covered engaging and shifting the ICD sleeve multiple times, from open to close and vice versa, with different techniques to ascertain baseline measurements to be used on the operation (Mauchien et al, 2021). All manipulations were done remotely by the engineer, relying solely on the measurements from the instrumented intervention tools to confirm, and separately verified using a borescope camera for demonstration to the operator as proof of a successful test. The equipment setup for demonstration to the operator can be seen in Figure 11, with remote monitoring and operating capabilities using a large touch screen interface for easier visualization during the test.



Figure 11—Workshop Equipment Setup for ICD Shifting SIT

The high expansion rotary mechanical cleaning tool performance was validated in a similar manner to the shifting tool. A retrieved section of used tubing, which had severe Barium Sulphate scaling throughout its entire length and circumference, was used for this test. The rotary cleaning tool was activated at 300-400RPM and conveyed slowly across the length of the scaled tubing. It was verified that the layer of scale was completely removed by the cleaning tool, all the way to the internal face of the tubing. Such consolidated debris was not expected in the shifting operation but regardless, the ability of the cleaning tool to remove even very hard debris from the completion proved that it would be effective to remove any debris present on the ICD assemblies prior to shifting.



Figure 12—Rotary Cleaning Tool Removing High Hardness Scale During SIT

Data and Results

The deployment of this combination of a high-strength, torque balanced wireline cable with a slim, intelligent tractor and instrumented intervention platform enabled extremely efficient ICD shifting in a sour, tortuous ERD well with Herringbone completion over 23,000ft deep. The production logging tools acquired high quality data to allow for a complete understanding of the flow contributions of each zone and the laterals. The subsequent precision mechanical intervention allowed for confident closure of the ten required ICDs to be followed by targeted acid stimulation of the Herringbone laterals. The following are the highlights of this operation:

- Maximum depth conveyed in a tortuous sour well: 23,100 ft
- Cumulative tractor distance in a single well: 39,000 ft
- Acquisition of production log data while simultaneously tracting down, increasing efficiency and yielding excellent production logging data to determine zonal flow contributions
- Tractor navigation through all ID changes from upper to lower completion, and efficient conveyance across the ICD assemblies in the lower completion using individual drive manipulation to navigate the variable IDs
- Highly efficient shifting of the target ICDs with all ten closed in under twelve hours

- Real-time monitoring of the shifting operation, through the array of integrated sensors, to allow for interpretation of the shifting ‘signature’ and a high degree of confidence to confirm complete and successful shifting of all ICDs
- Targeted acid stimulation of the Herringbone laterals after shifting to significantly increase the well productivity

Case Study

The operator drilled and completed a well with a new type of completion design, with four Herringbone laterals from the main wellbore in the horizontal section. The objective of designing and completing a well with such construction was to maximize the well productivity, and formation contact, from a single slot at surface. This increased performance would allow the operator to yield higher production rates from a reduced number of wells in the field.

The production logging survey was performed to assess the downhole flow profile and determine the contribution of individual ICDs to the overall production. The survey was conducted using two choke settings, both of which exhibited similar flow profiles. However, the 80% choke setting provided superior data quality compared to previous surveys conducted at lower choke settings. The results indicated that all target ICDs, except for ICD #19, were actively contributing to production. A post-production survey has been planned to verify the shifting of all ICDs and evaluate the effectiveness of the operation.

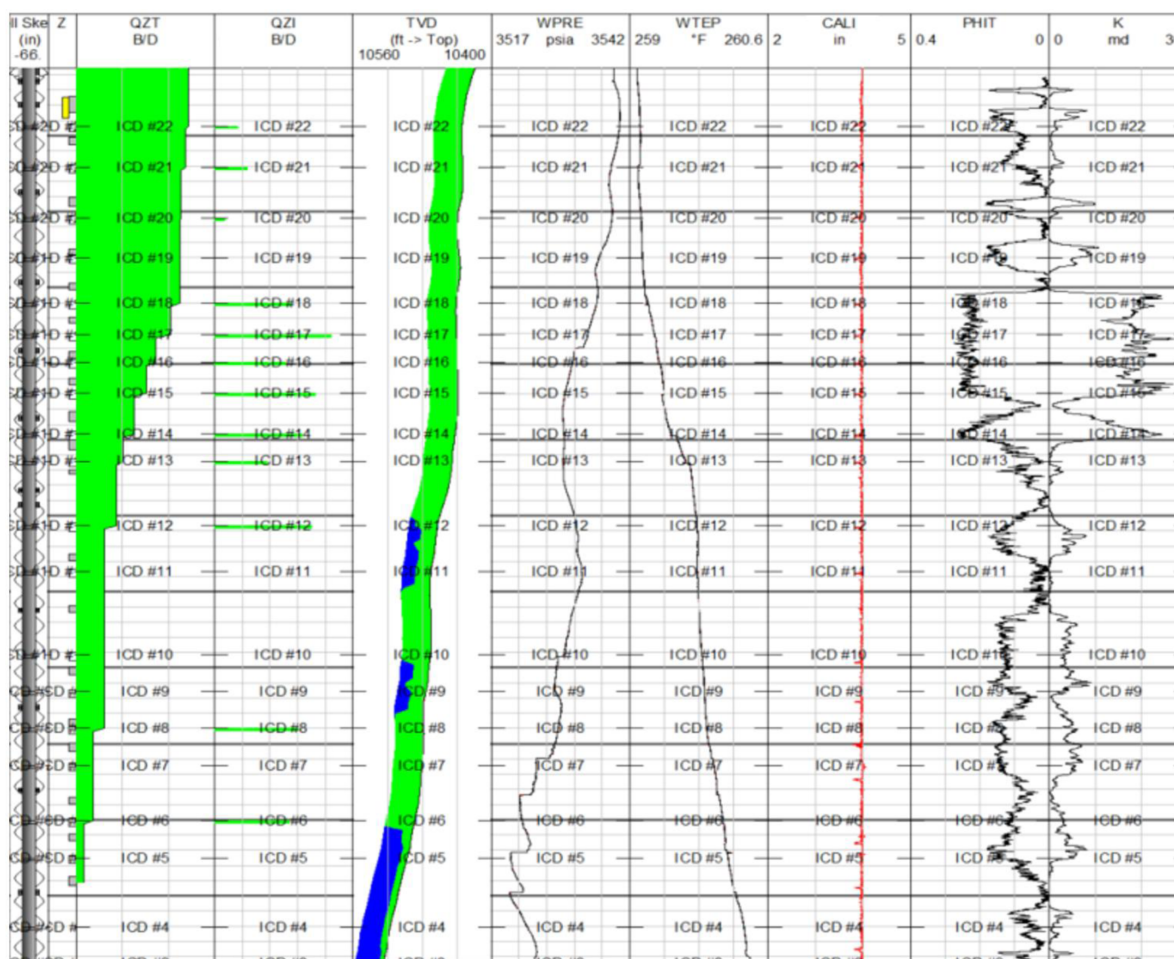


Figure 13—Production Logging Results for Decision on ICDs to Close

Once the target ICDs had been determined, the same combination of torque-balanced sour-service cable and intelligent extreme wireline tractor was used to convey the instrumented shifting tools into position. Starting with the deepest target ICD, the tools were conveyed just below the ICD assembly and the shifting tool was activated in a low pressure, seeking mode. The tools were then picked up using the wireline winch to latch onto the ICD sleeve shifting profile, observed by an overpull of several hundred pounds force on the downhole tension sensor at the cable head. The low-pressure mode allows for some compliance in the shifting tool mechanism to successfully pass any other ID changes in the assembly and a ‘signature’ of the seeking pressure is acquired in real-time during the latching process (Mauchien et al, 2021). Once the shifting tool key was engaged with the sleeve, the anchor was set to fix the tools in position, the shifting key pressure was increased to ensure a strong latch, and the linear actuator was retracted to move the sleeve upwards to close it. During the closing process multiple sensors integrated into the intervention platform are monitored in real-time to ensure that the sleeve has been moved to the desired position. The diameter and opening force sensors in the anchor module confirm that it is in the desired section of tubing with a known ID, and a constant anchor force reading indicates that there was no slippage during the shifting process. The travel distance and expected force to move the ICD sleeve are documented in the relevant technical drawings and used as a reference point. In this case, the expected force to move the sleeve was around 1,200lbf and the travel distance was 1.7". The linear displacement and force sensors of the linear actuator were monitored to verify that the expected force and travel distance were observed. This additional measurement of linear displacement is invaluable for shifting ICDs to provide additional assurance of complete movement. Finally, the hydraulic opening pressure of the shifting tool was monitored to ensure that the keys remained firmly latched with the ICD sleeve throughout the shifting process. Figure 14 shows an example of the shifting process for one of the ICDs closed during this operation. At the initial point of engagement with the lower sleeve profile, the linear actuator was in a starting position of 10.0". Following a successful latch, the shifting tool opening pressure was increased to around 2000psi and the sleeve was shifted exactly 1.7" to end at a stroke position of 8.3", matching precisely the technical drawing of the ICD. At this point a spike in the shifting force was observed, indicating that the sleeve had reached its maximum travel and was closed. As confirmation the shifting tool was closed then re-positioned to the same starting point with the liner actuator at 10.0", keeping the anchor engaged at the same point throughout, then re-opened and the linear actuator retracted once again to engage the shifting tool with the sleeve at 8.3" once more. The process of latching, shifting and confirming was repeated in the same way for all ten of the target ICDs in this well. The log signatures could all be overlaid on top of each other to confirm that they matched exactly, removing any doubts as to the final position of the ICD sleeves.

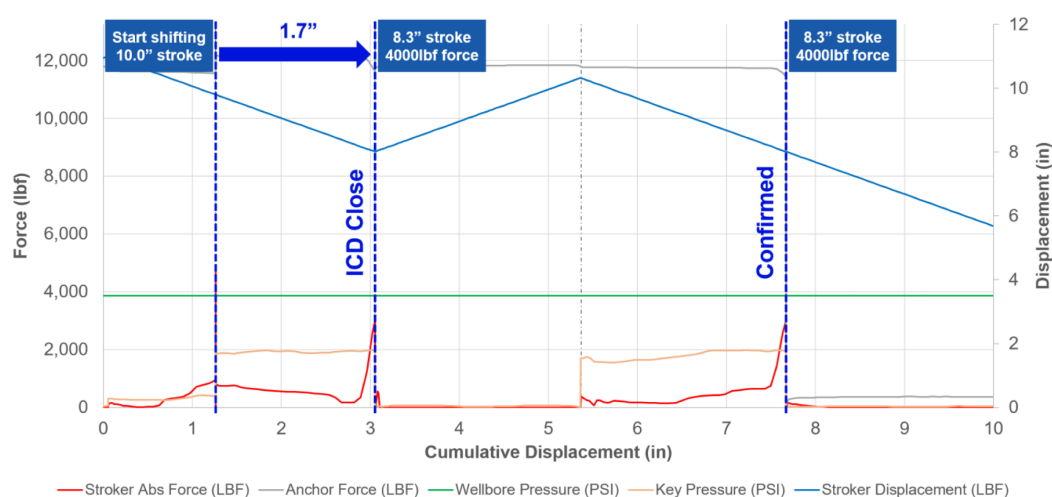


Figure 14—Real-time Precision Measurements of ICD Manipulation

During the latching process of the ninth ICD in the sequence, when taking the overpull with the winch, it was observed that the sleeve had partially moved. This was identified once the shifting process was started, the travel distance was observed as just 1.0" instead of 1.7". At this stage, it could have been concluded that either the sleeve was not able to move fully due to some obstruction or that it had been inadvertently moved with the force applied by the winch when latching. To remove any doubt and confirm which scenario had occurred, the ICD sleeve was manipulated in multiple cycles of closing and re-opening. In the subsequent opening and closing cycles, it was clearly observed that the sleeve could be moved up and down by the exact distance of 1.7" as expected. With the final sleeve manipulation being 1.7" during closure and having a signature which matched exactly the other nine ICDs. This again left no doubts that the sleeve had been left in the closed position. Figure 15 shows the last two stages of this multi-step process with one cycle of opening, one cycle of closing and then confirmation. All the ICD shifting log signatures were analyzed and compared in real-time during the operation. Accurate signature matching across multiple integrated sensors concluded with a high degree of confidence that all 10 ICD sleeves had been closed before pulling out of hole to conclude the wireline operation and allow the subsequent targeted acid stimulation of the Herringbone laterals to be completed.

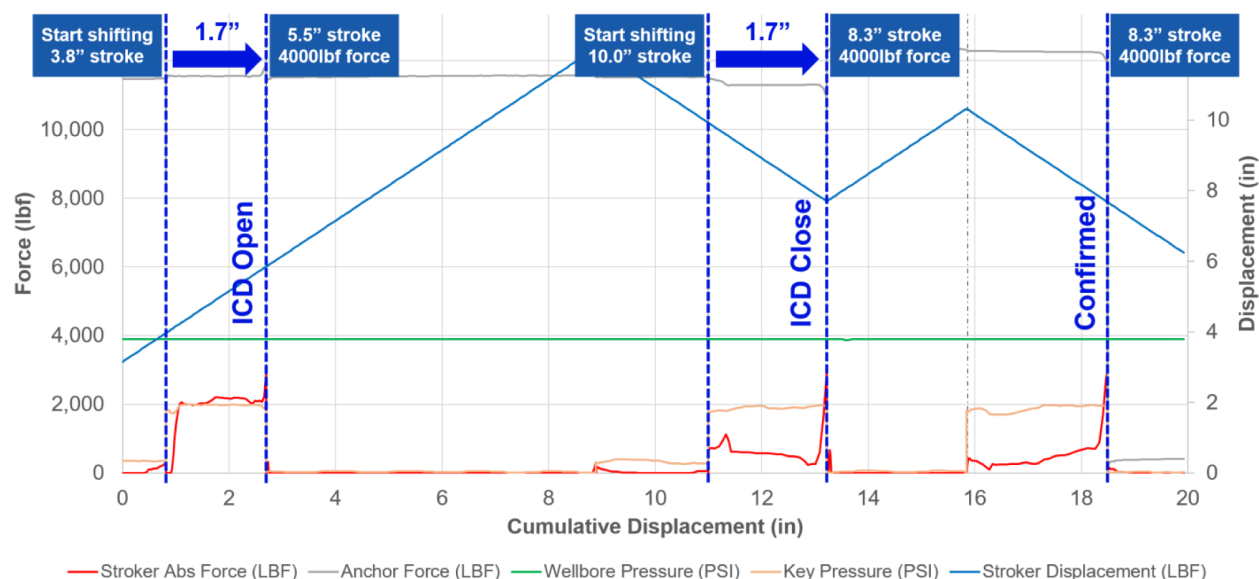


Figure 15—Real-time Precision Measurements of ICD Manipulation

Stimulation Treatment

The matrix stimulation treatment was performed combining Flexible Single-Phase Retarded Acid (FSPRA) systems and viscoelastic-based diverting agents (VBDA). This innovative approach optimizes stimulation efficiency for this type of complex well' architecture. Laboratory tests demonstrated the superior retardation and deeper penetration of FSPRA systems compared to conventional acids, and the combination with VBDA enhanced stimulation placement of the Herringbone-type legs along the main lateral after the ICD shifting, doubling the Productivity Index in the more marginal reservoir sections.

The synergistic use of these advanced acidizing techniques and the unique Herringbone Multilateral design significantly improved productivity and recovery in medium-to-low permeability carbonates. This methodology offers a robust, efficient strategy to address the challenges of complex geometries, underscoring its potential for broader application in maximizing reservoir performance.

Conclusions

The case study presented in this paper has proved that efficient, reliable, wireline deployed intervention to mechanically clean and shift ICDs in extreme, tortuous sour wells is possible. The unique combination of the slim OD, intelligent, instrumented sour service tractor and intervention platform combined with the unique torque-balanced, high-strength, sour service wireline cable can overcome the multiple legacy challenges associated with such interventions. Key achievements included:

- Development of bespoke shifting keys and a custom, slim OD high expansion rotary wireline mechanical cleaning tool
- Achieving target depth to complete multiple downhole operations, navigating through complex completions in a challenging, tortuous trajectory Herringbone completion well with an intelligent wireline tractor at over 23,000ft depth
- Deployment of wireline intervention tools via a high-strength, high-power, torque-balanced wireline cable minimizing conventional risks of raised armors and potential stuck cable scenarios
- Utilizing a comprehensive suite of sour service wireline tractor, intervention tools and cable thereby eliminating the risk of corrosion-related downhole equipment failures
- Effective management of well integrity through improved sealing capability between wireline PCE and polymer-filled cable even in a high-pressure gas environment
- Identification of compartments requiring shut-off with an advanced multi-array production logging tool
- Efficient closure of ten ICDs in under twelve hours to allow for targeted acid stimulation of the Herringbone laterals
- Real-time analysis of the shifting signature from the multitude of sensors integrated into the instrumented shifting tools with immediate confirmation and extremely high degree of confidence of successful sleeve closure
- From the Stimulation side, this study demonstrates the effectiveness of combining FSPRA systems with VBDA to optimize stimulation treatments in Herringbone Multilateral Horizontal Wells with ICDs, offering an efficient strategy to improve performance and maximize reservoir recovery in medium-to-low permeability carbonates.

The rapid development of custom solutions to augment the existing sour service, instrumented intervention platform has resulted in a step-change in the efficiency and reliability of mechanical intervention for the operator. A comprehensive, wireline deployed production logging and mechanical intervention platform enabled rapid completion of the required downhole operations to clean and shift ICDs even in the most challenging well conditions. The success of these operations demonstrated the value that instrumented electric line mechanical intervention tools bring. The ability to utilize readings from multiple independent sensors to confirm successful shifting, as well as precisely controlled linear force and displacement, enabled an efficient operation without any doubts of incomplete shifting. This has allowed the operator to efficiently perform powerful but lightweight mechanical interventions on wireline and opens the door for further expansion of such operations.

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